

Total No. of Questions : 8]

SEAT No. :

PA-1417

Jan-2023

[Total No. of Pages : 2

[5926]-33

T.E. (Chemical)

TRANSPORT PHENOMENA

(2019 Pattern) (Semester - II) (309350)

Time : 2½ Hours]

[Max. Marks : 70]

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Derive the expression of molar flux, concentration profile and average concentration for diffusion with homogeneous chemical reaction. [12]

b) Diffusivity of gas-pair oxygen-carbon tetrachloride is determined by observing steady state evaporation of carbon tetrachloride. The distance between CCl_4 liquid level and top of tube is 17.1 cm. The total pressure on the system is 755 mm-Hg and temperature is 0°C . Vapor pressure of CCl_4 at this temperature is 33 mm Hg. Cross-sectional area of tube is 0.82 cm^2 . If it is found that 0.0208 cm^3 of CCl_4 evaporate in 10-hour period, what is diffusivity of gas-pair $\text{CCl}_4 - \text{O}_2$? (Density of $\text{CCl}_4 = 1.59 \text{ g/cm}^3$) [6]

OR

Q2) a) Consider a natural gas mixture contained in a pyrex tube with radius as R_1 and R_2 . Obtain an expression for the rate at which helium will leak through the tube. [12]

b) Explain stepwise procedure to solve mass transfer problems. [6]

Q3) a) Use Navier Stoke's equation of motion to derive Hagen Poiseuille equation. [12]

b) What do you mean by partial time derivative, total time derivative and substantial time derivative? [5]

OR

P.T.O.

- Q4) a) Derive Newton's second law of motion. [12]
b) What do you mean by equation of change for isothermal system? [5]

- Q5) a) Derive Blake Kozeny equation for laminar flow of fluid through packed bed. [12]
b) Derive expression for friction factor and Reynold number for laminar flow through fluid through pipe. [6]

OR

- Q6) a) Derive Burke Plummer equation for turbulent flow of fluid through packed bed. [12]
b) Explain macroscopic mass balance equation. [6]
- Q7) a) Explain Martinnelli's analogy. [9]
b) Explain Reynold analogy. [8]

OR

- Q8) a) Explain Chilton and Colburn analogy. [9]
b) A spherical water droplet, 0.05 cm in diameter is falling at velocity of 215 cm/sec through dry, still air at 1 atm pressure. Estimate instantaneous rate of evaporation from the drop if drop surface is at 21°C and air at 60° C. Vapor pressure of water at 21°C is 0.0247 atm. Assume pseudo steady state condition and $k_{xm} = 1.35 \times 10^{-3} \text{ mol s}^{-1} \text{ cm}^{-2}$. [8]

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Total No. of Questions : 8]

P802

June 2022

SEAT No. :

[Total No. of Pages : 2

[5870]-1120

T.E. (Chemical)

TRANSPORT PHENOMENA

(2019 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Derive the expression of molar flux, concentration profile and average concentration for diffusion through stagnant gas. [12]

- b) Calculate the mass flux of benzene through a layer of air 10 mm thickness at 25 °C and 200 kN/m² (total pressure), partial pressure of benzene is 6×10^3 N/m² at the left side of the layer and 1 kN/m² at the right side. The mass diffusivity at this temperature and pressure is 4.4×10^{-6} m²/s? [6]

OR

Q2) a) Derive the expression of molar flux and concentration profile for diffusion with heterogeneous (instantaneous and dimerization) chemical reaction. [12]

- b) What is rate of evaporation of chloropicrin into air at 298 K? Total pressure = 770 mm Hg, diffusivity = 0.088 cm²/sec, vapor pressure = 23.81 mm Hg, $Z_2 - Z_1 = 11.14$ cm, density of chloropicrin = 1.65 g/cm³, $S = 2.29$ cm². [6]

Q3) a) Derive Navier Stoke's equation of motion. [12]

- b) Give significance of different types of derivatives. [5]

OR

Q4) a) Derive equation of continuity in three dimensional form. [12]

- b) Derive dimensionless form of equation of change. [5]

P.T.O.

Q5) a) Derive Ergun equation for flow through packed bed. [12]

b) Show that for laminar flow of fluid through the tube $f = 16/Re$. [6]

OR

Q6) a) Derive expression of fanning friction factor. [12]

b) What pressure gradient is required to cause N, N-diethylamine to flow in a horizontal smooth tube of inside diameter 3 cm at a rate of 1.1 lit/s. Density of diethylamine is 935 kg/m³ and viscosity is 1.95 cp. (Assume friction factor $f = 0.0063$). [6]

Q7) a) Explain Chilton and Colburn analogy. [9]

b) A spherical water droplet, 0.05 cm in diameter is falling at velocity of 215 cm/sec through dry, still air at 1 atm pressure. Estimate instantaneous rate of evaporation from the drop if drop surface is at 21°C and air at 60°C. Vapor pressure of water at 21°C is 0.0247 atm. Assume pseudo steady state condition and $k_{xm} = 1.35 \times 10^{-4}$ mol s⁻¹cm⁻². [8]

OR

Q8) a) Explain Martinelli's analogy. [9]

b) Explain co-relation of binary mass transfer coefficient in one phase at low mass transfer rates. [8]



[5870]-1120

T.E. (Chemical)

TRANSPORT PHENOMENA

(2019 Pattern) (Semester - II)

Time : 2½ Hours]

Instructions to the candidates :

[Max. Marks : 70

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Need diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Derive the expression of molar flux, concentration profile and average concentration for diffusion through stagnant gas. [12]

b) Calculate the mass flux of benzene through a layer of air 10 mm thickness at 25 °C and 200 kN/m² (total pressure), partial pressure of benzene is 6×10^3 N/m² at the left side of the layer and 1 kN/m² at the right side. The mass diffusivity at this temperature and pressure is 4.4×10^{-6} m²/s? [6]

OR

Q2) a) Derive the expression of molar flux and concentration profile for diffusion with heterogeneous (instantaneous and dimerization) chemical reaction. [12]

b) What is rate of evaporation of chloropierin into air at 298 K? Total pressure = 770 mm Hg, diffusivity = 0.088 cm²/sec, vapor pressure = 23.81 mm Hg, $Z_2 - Z_1 = 11.14$ cm, density of chloropierin = 1.65 g/cm³, $S = 2.29$ cm². [6]

Q3) a) Derive Navier Stoke's equation of motion. [12]

b) Give significance of different types of derivatives. [5]

OR

Q4) a) Derive equation of continuity in three dimensional form. [12]

b) Derive dimensionless form of equation of change. [5]

P.T.O.

Q5) a) Derive Ergun equation for flow through packed bed. [12]

b) Show that for laminar flow of fluid through the tube $f = 16/Re$. [6]

OR

Q6) a) Derive expression of laminar friction factor. [12]

b) What pressure gradient is required to cause N,N-diethylamine to flow in a horizontal smooth tube of inside diameter 3 cm at a rate of 1.1 lit/s. Density of diethylamine is 935 kg/m³ and viscosity is 1.95 cp. (Assume friction factor $f = 0.0063$). [6]

Q7) a) Explain Chilton and Colburn analogy. [9]

b) A spherical water droplet, 0.05 cm in diameter is falling at velocity of 215 cm/sec through dry, still air at 1 atm pressure. Estimate instantaneous rate of evaporation from the drop if drop surface is at 21 °C and air at 60 °C. Vapor pressure of water at 21 °C is 0.0247 atm. Assume pseudo steady state condition and $k_{m} = 1.35 \times 10^{-3}$ mol s⁻¹ cm⁻². [8]

OR

Q8) a) Explain Martinelli's analogy. [9]

b) Explain co-relation of binary mass transfer coefficient in one phase at low mass transfer rates. [8]

T.E. (Chemical Engineering)
TRANSPORT PHENOMENA
(2019 Pattern) (Semester - II) (309350)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, and Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Derive the expression of molar flux, concentration profile and average concentration for diffusion with homogeneous chemical reaction. [12]

b) What is rate of evaporation of chloropicrin into air at 298 K? [6]

Total pressure = 770 mm Hg, diffusivity = $0.088 \text{ cm}^2/\text{sec}$, vapor pressure = 23.81 mm Hg, $Z_2 - Z_1 = 1.14 \text{ cm}$, density of chloropicrin = 1.65 g/cm^3 , $S = 2.29 \text{ cm}^2$.

OR

Q2) a) The solute HCl is diffusing through a thin film of water 2.0mm. The concentration of HCl at point 1 is 12.0 wt% (density = 1061 kg/m^3), and at point 2 is 6.0 wt% (density = 1030 kg/m^3).

Diffusivity of HCl in water is $2.5 \times 10^{-9} \text{ m}^2/\text{s}$. Calculate the flux of HCl in $\text{kg-mole/m}^2\cdot\text{s}$

[6]

b) Consider a natural gas mixture contained in a pyrex tube with radius as R_1 and R_2 . Obtain an expression for the rate at which helium will leak through the tube.

[12]

P.T.O.

Q3) a) Use Navier Stoke's equation of motion to derive Hagen Poiseuille Equation. [12]

b) Give significance of different types of derivatives. [5]

Q4) a) Derive Newton's second law of motion. [12]

b) Derive dimensionless form of equation of change. [5]

Q5) a) Derive Blake Kozeny equation for laminar flow of fluid through packed bed. [12]

b) Derive expression for friction factor and Reynold number for laminar flow through fluid through pipe. [6]

Q6) a) Derive Burke Plummer equation for turbulent flow of fluid through packed bed. [12]

b) Explain macroscopic mass balance equation. [6]

Q7) a) Explain Martinelli's analogy. [9]

b) Explain Reynold analogy. [8]

[6003] - 322

2

Q8) a) Explain co-relation of binary mass transfer coefficient in one phase at low mass transfer rates. [9]

b) A spherical water droplet, 0.05 cm in diameter is falling at velocity of 215 cm/sec through dry, still air at 1 atm pressure. Estimate instantaneous rate of evaporation from the drop if drop surface is at 21° C and air at 60° C. Vapor pressure of water at 21° C is 0.0247 atm. Assume pseudo steady state condition and $D_{am} = 1.35 \times 10^{-5}$ mol s⁻¹ cm⁻². [8]

[6003] - 322

3

Total No. of Questions : 8]

P7532

SEAT No. : 1910015020

[Total No. of Pages : 2

[6180]-40

T.E. (CHEMICAL)

TRANSPORT PHENOMENA

(2019 Pattern) (Semester - II) (309350)

(Backlog 2023)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Derive the expression of molar flux, concentration profile and average concentration for diffusion with homogeneous chemical reaction. [12]

b) Explain mass balance equation for microscopic systems. [6]

OR

Q2) In studying rate of leakage of substance A from solid particles by solvent B, rate controlling step is diffusion from A from particle surface through liquid film into a main liquid stream. Solubility of A in B is C_{A0} mol/cm³ and concentration in the main stream is $C_{A\delta}$ mol/cm³. [18]

- a) Obtain differential equation for C_A as a function of Z .
- b) Show that in absence of chemical reaction in the liquid phase, the concentration profile is linear.
- c) Show that rate of leaching is then given by.

$$N_{AZ} = D_{AB} \frac{C_{A0} - C_{A\delta}}{\delta}$$

Q3) Derive Euler's equation of motion. [17]

OR

P.T.O.

Q4) a) Explain different types of derivatives used in deriving equation of change. [5]

b) Use Navier-Stokes equation of motion to determine velocity distribution for laminar flow of Newtonian fluid through vertical pipe. [12]

Q5) a) Show that for laminar flow of fluid through the tube $f = 16/Re$. [6]

b) An incompressible fluid flows turbulently in a circular tube of cross sectional area S_1 which empties into a large tube of cross sectional area S_2 . Use macroscopic balances and derive expression for pressure rise and friction loss due to sudden expansion. [12]

OR

Q6) a) Derive Blake Kozeny equation for laminar flow of fluid through packed bed. [12]

b) What pressure gradient is required to cause N, N-diethylamine to flow in a horizontal smooth tube of inside diameter 3 cm at a rate of 1.1 lit/sec: Density of diethylamine is 935 kg/m^3 and viscosity is 1.95 cp. Assume factor of ' f ' = 0.0063. [6]

Q7) a) Explain co-relation of binary mass transfer coefficient in one phase at low mass transfer rates. [9]

b) A spherical water droplet 0.05 cm in diameter is falling at velocity of 215 cm/sec through dry, still air at 1 atm pressure. Estimate instantaneous rate of evaporation from the drop if drop surface is at 21°C and air at 60°C . Vapor pressure of water at 21°C is 0.0247 atm. Assume pseudo steady state condition and $k_m = 1.35 \times 10^{-3} \text{ mol S}^{-1} \text{ cm}^{-1}$. [8]

OR

Q8) a) Explain Chilton and Colburn analogy. [9]

b) Explain Martinnelli's analogy. [8]